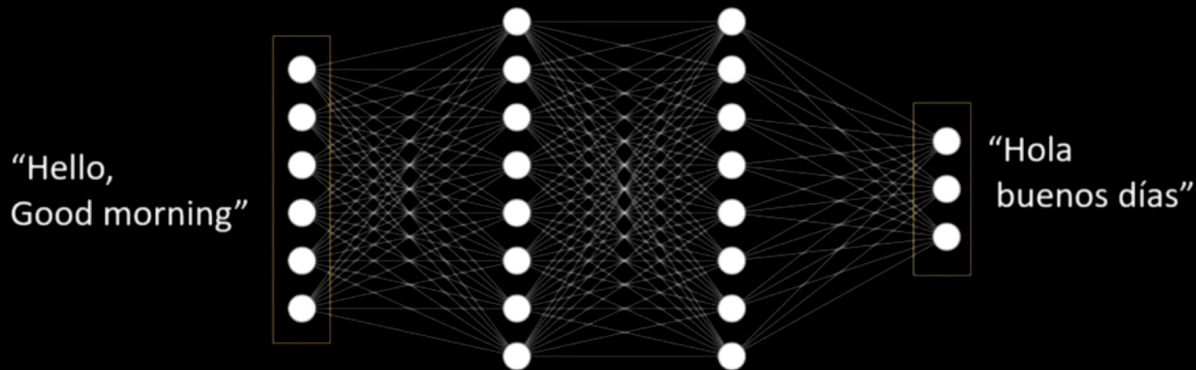


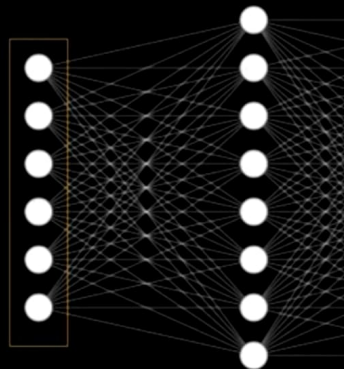
Problem 1: Fixed number of input and output neurons



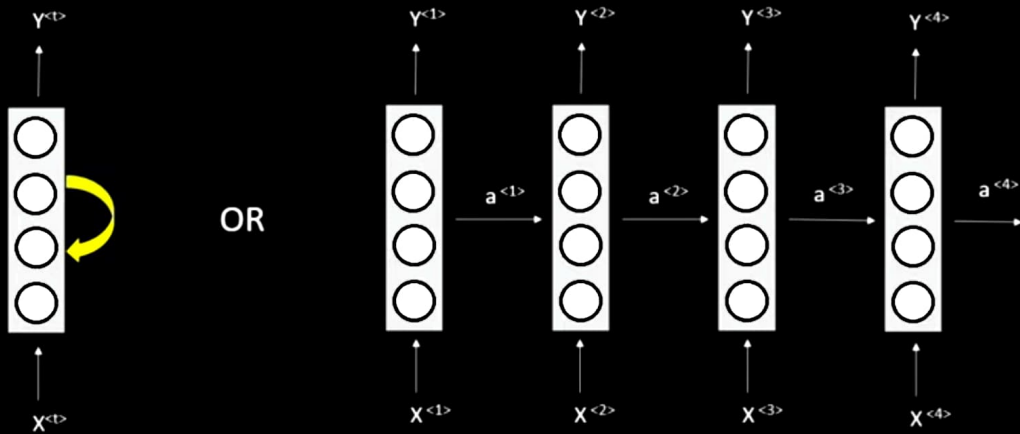
Kelly	worked	at	Google	in	San	Francisco	0	0	0
Kelly	is	a	Software	Engineer	at	Google	in	San	Francisco
Kelly	is	very	talented	0	0	0	0	0	0

Problem 2: Does not take input sequence into consideration

English	↔	Spanish
Kelly worked at Google in San Francisco	×	Kelly trabajó en Google en San Francisco.
At Google, in San Francisco, Kelly worked.	×	En Google, en San Francisco, Kelly trabajó.



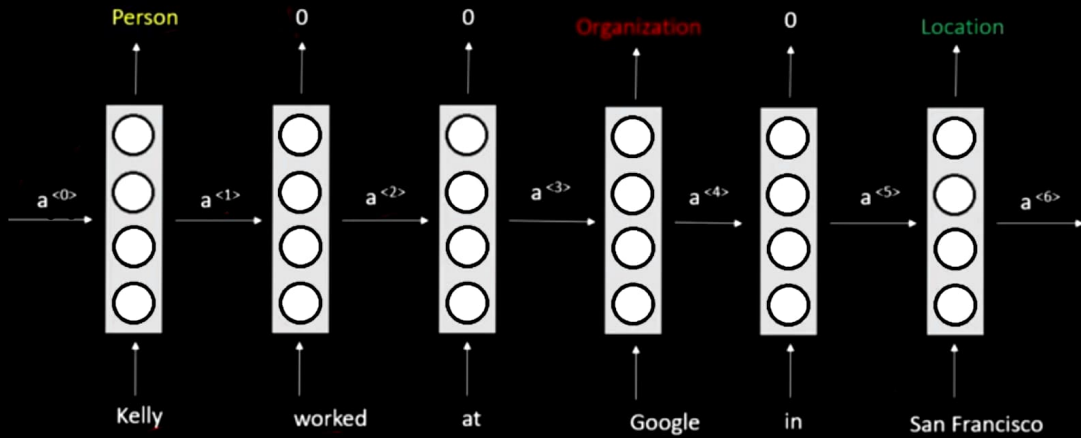
RNN



Named Entity Recognition

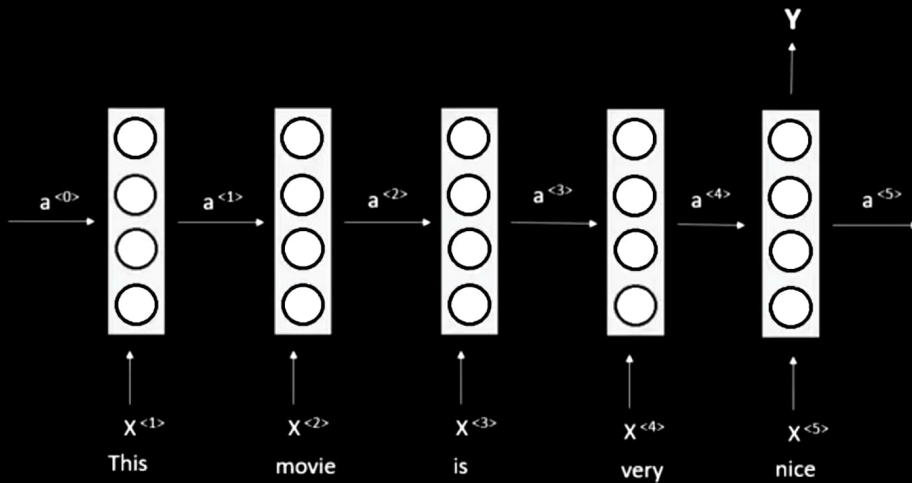
Person **Organization** **Location**
└──┬──┘ └──┬──┘ └──┬──┘
Kelly worked at Google in San Francisco.

Location **Date**
└──┬──┘ └──┬──┘
"Hey, show me the best places to visit in Mumbai during Jan 2022"

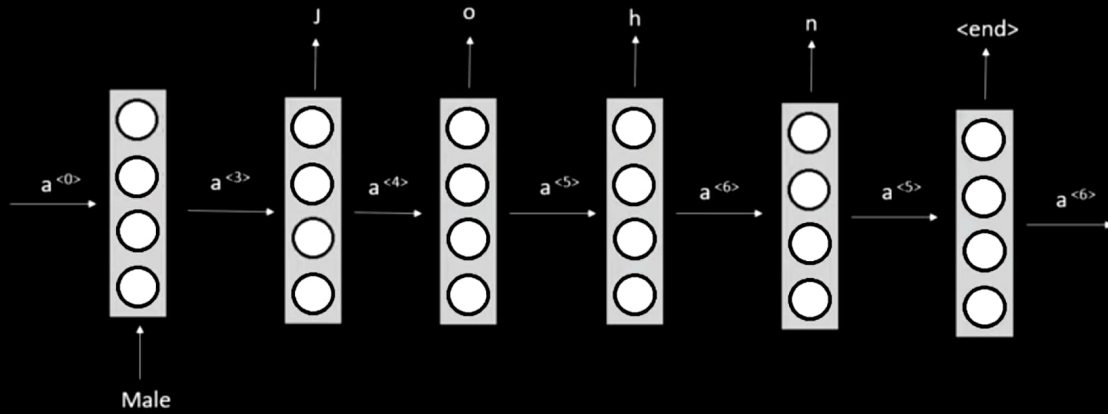


Kelly worked at Google in San Francisco.

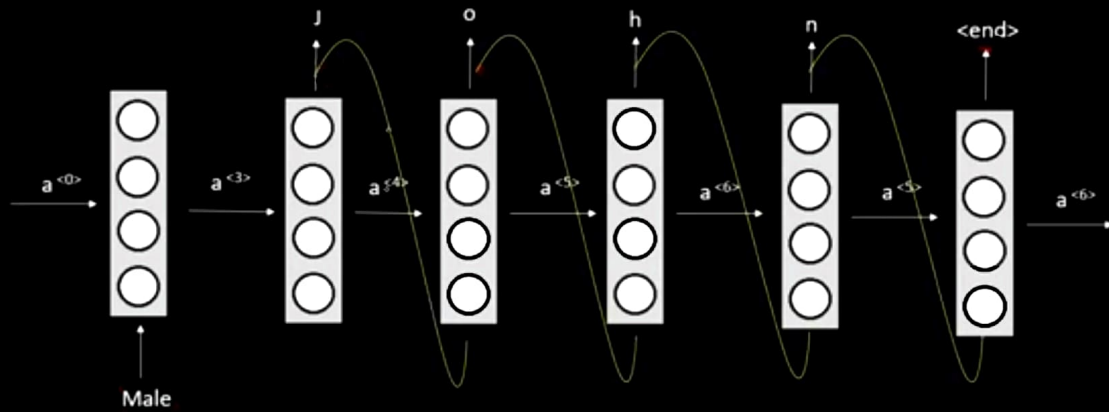
"This movie is very nice!"



Baby Name Generation



Baby Name Generation



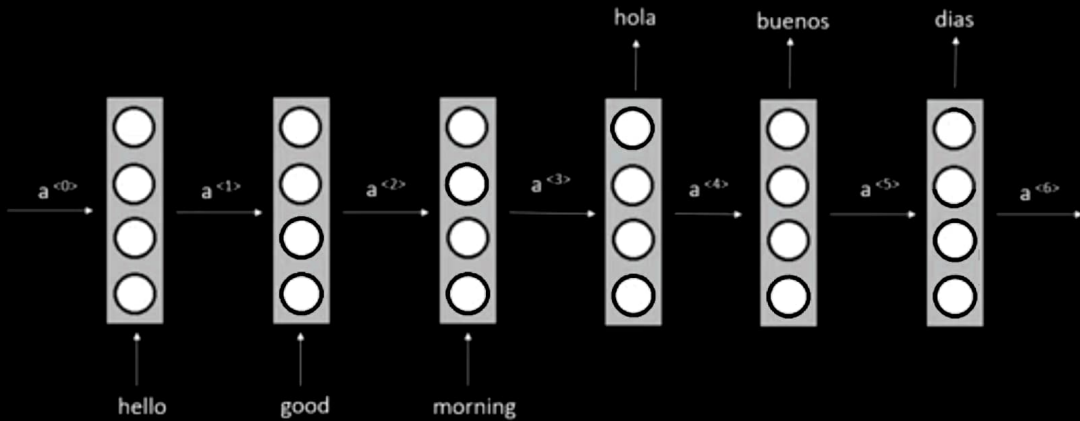
English

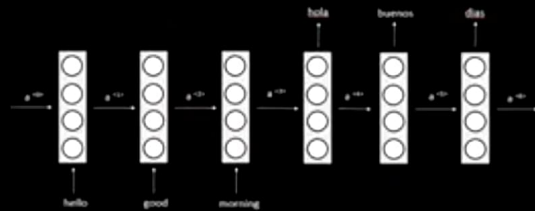
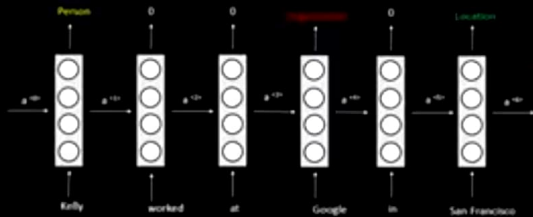
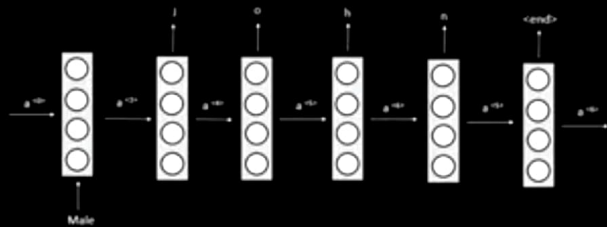
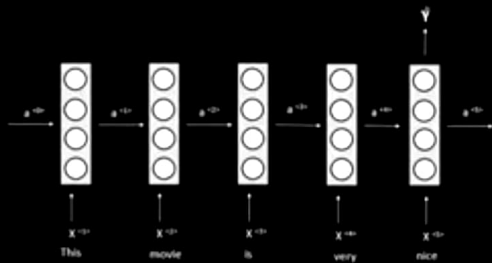
Spanish

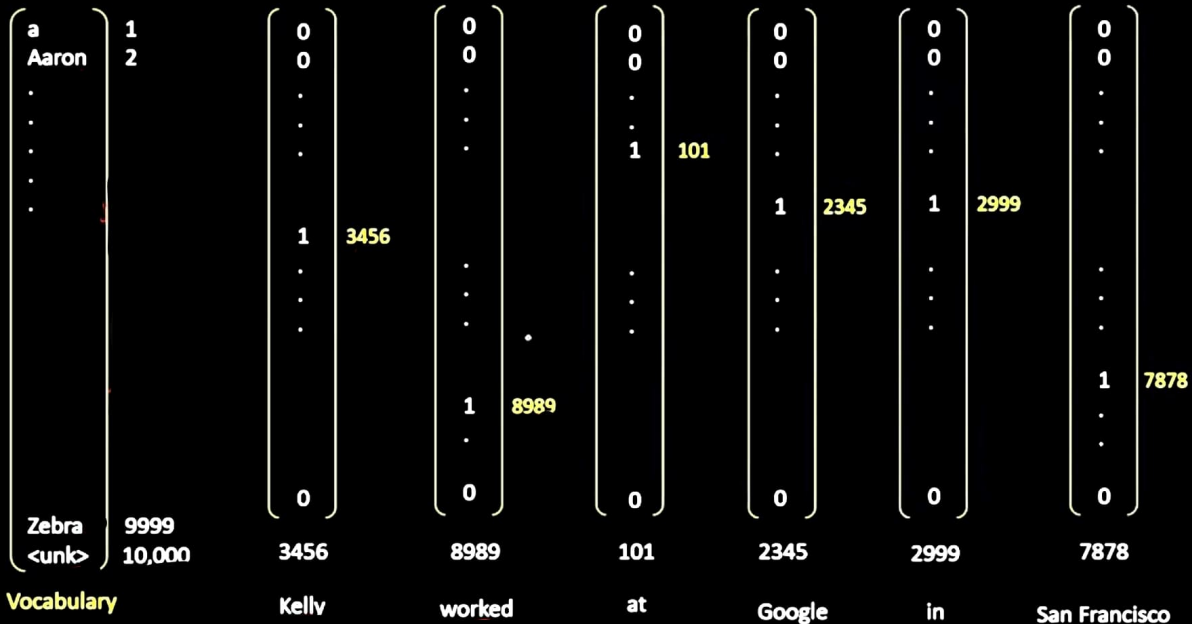
hello good morning

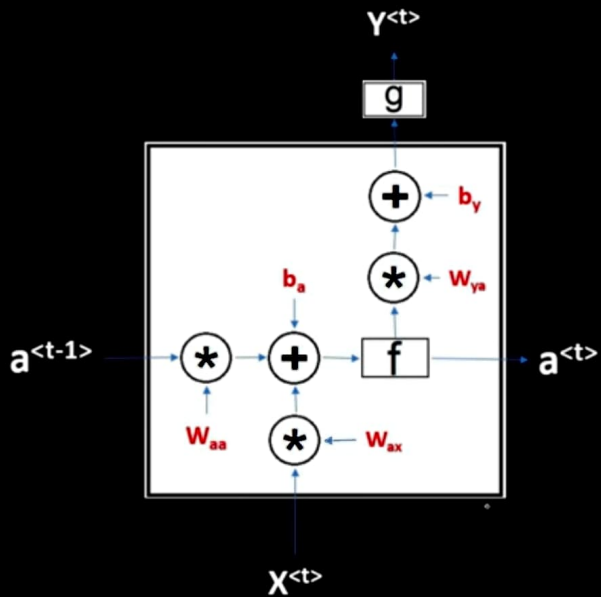
×

Hola buenos dias









For ANN

$$a = \text{function}(W * a_{t-1} + B)$$

For RNN

$$a = \text{function}(W_{aa} * a_{t-1} + W_{ax} * X + b_a)$$

$$Y = \text{softmax}(W_{ya} * a_t + b_y)$$

Advantages of RNN:

- LSTM (Long short term memory)
- i) Useful in Time Series Prediction
 - remembers info through time
- ii) Used to extend pixel neighborhood.
 - when used with Convolutional Layer.

Disadvantages of RNN:

- i) Problems like Gradient Vanishing
- ii) Training is difficult.
- iii) Not very efficient in long sequences processing.

RNN

Depends on
Previous
moment
both
Current
ip and
Previous
op.

Feed Forward NN

ip are fed to NN
and transformed into
op.

used in classification.

Prediction does not
depend on predictions
in previous moment.